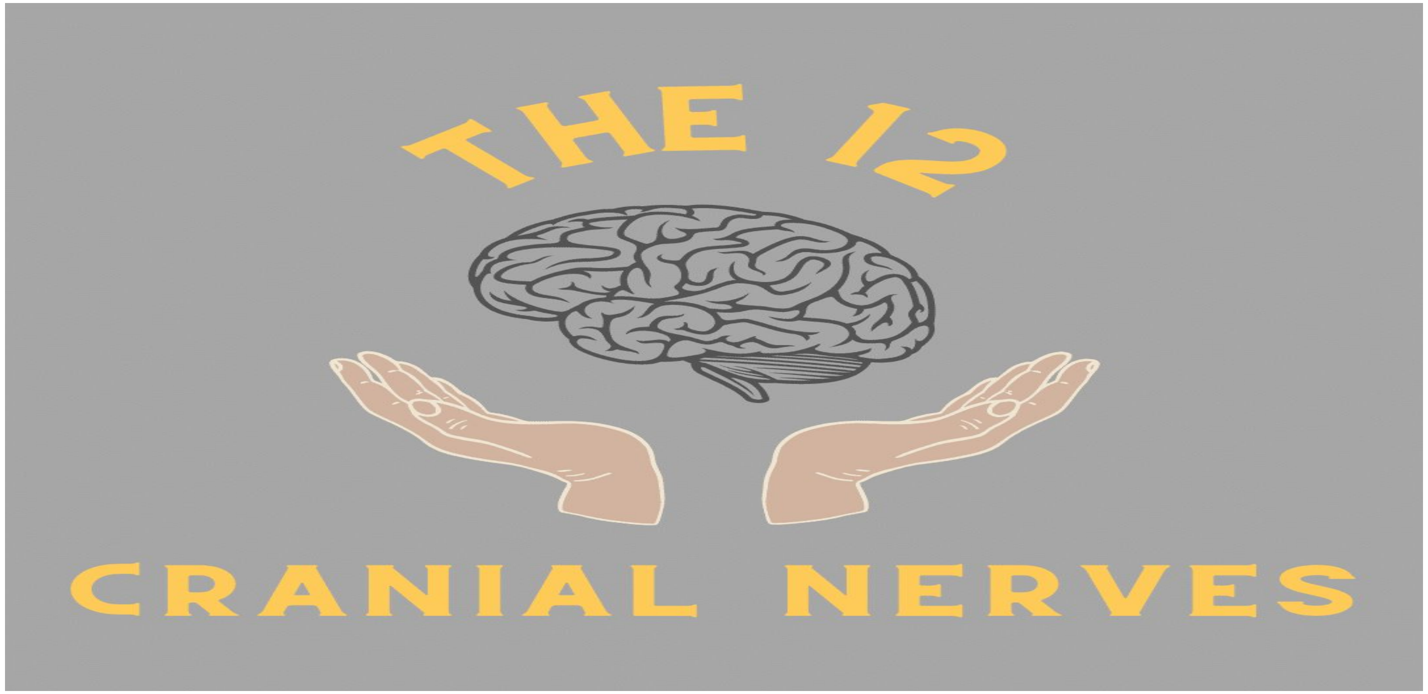
CRANIAL NERVES LESION AND ASSOCIATED DEFICITS



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LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- List all 12 cranial nerves in order.
- Classify each cranial nerve as sensory, motor or mixed.
- Describe the primary function of each nerve(e.g. smell, vision, eye movement, facial sensation, swallowing).
- Relate each nerve to its target structures.
- Identify key sign & symptoms when a specific cranial nerve is damaged.
- Use symptoms to localize neurological lesion.
- Perform step by step clinical testing of each nerve.

What are nerves?

Nerves are the bundles of axons in the peripheral nervous system that transmit electrochemical signals throughout the body.

CLASSIFICATION OF NERVES

Nerves can primarily be classified in two ways:

1. Functional classification
2. Structural classification

FUNCTIONAL CLASSIFICATION OF NERVES

According to the function, nerves can be classified into:

1. Sensory nerves
2. Motor nerves
3. Mixed nerves

1. SENSORY NERVES(AFFERENT)

Sensory nerves carry sensory information(touch, taste, smell, pain, temperature) from sense organs of the body to the brain and spinal cord(CNS).

CONT..

2. MOTOR NERVES(EFFERENT)

Motor nerves carry commands from the brain and spinal cord to muscles and glands enabling their movement & function.

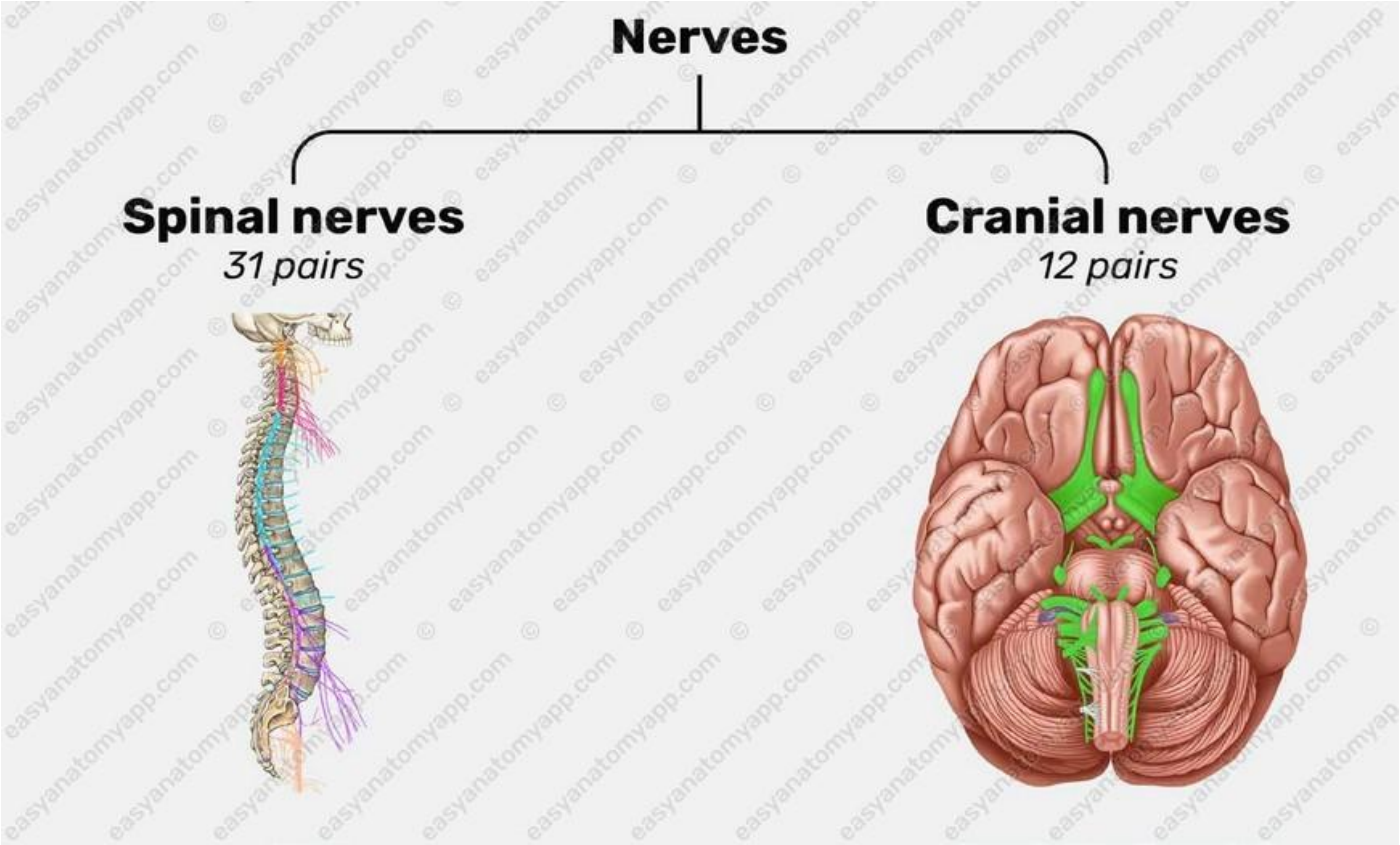
3. MIXED NERVES(BOTH)

These nerves contain both the sensory & motor fibres allowing them to conduct signals in both directions between the CNS & PNS.

STRUCTURAL CLASSIFICATION OF NERVES

Anatomically, nerves can be classified into:

- Cranial nerves
- Spinal nerves


Nerves
Spinal nerves
31 pairs
Cranial nerves
12 pairs

CONT..

1. ***CRANIAL NERVES***

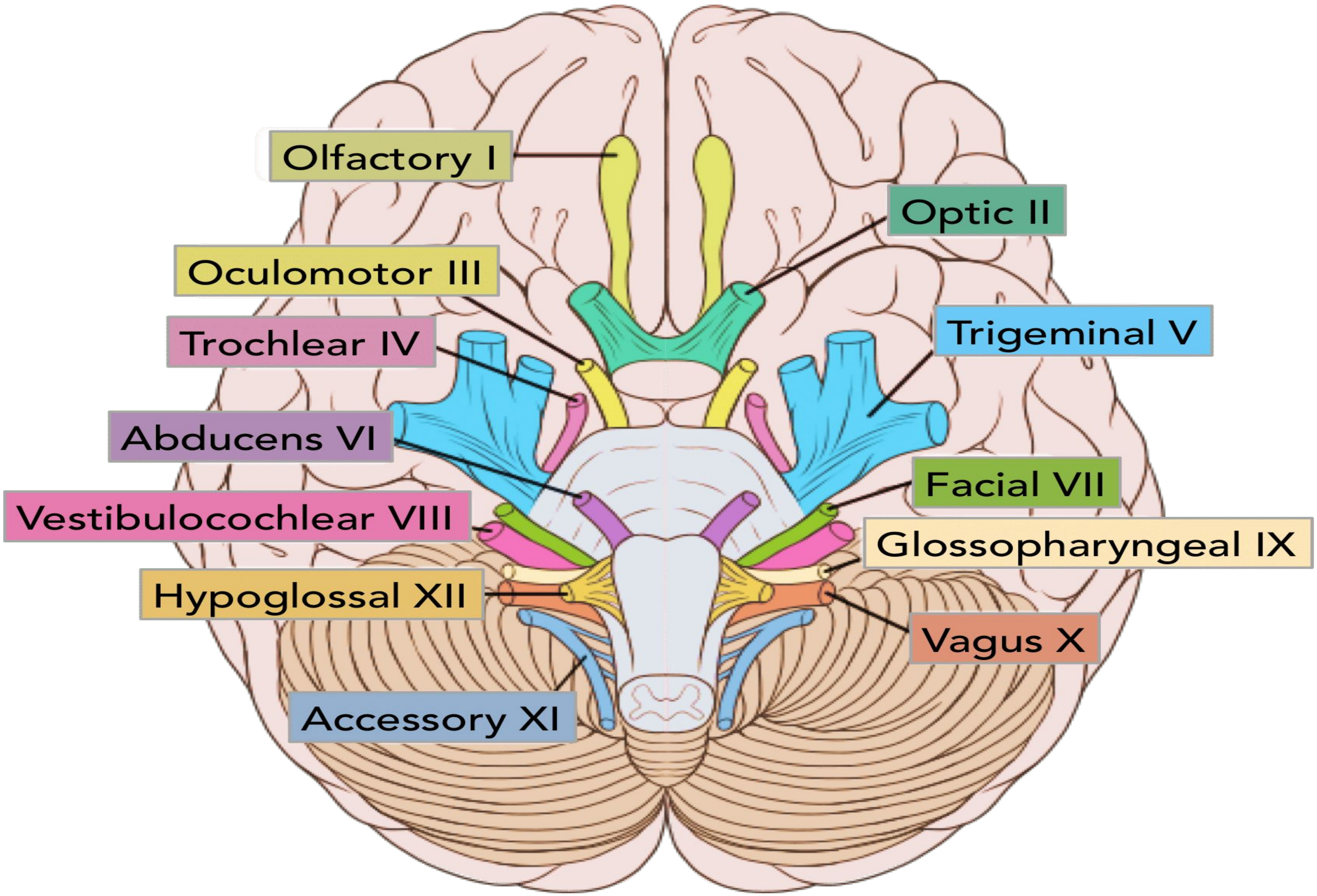
Cranial nerves are the 12 pairs of nerves originating directly from the brain that control head, neck and specialized sensory functions.

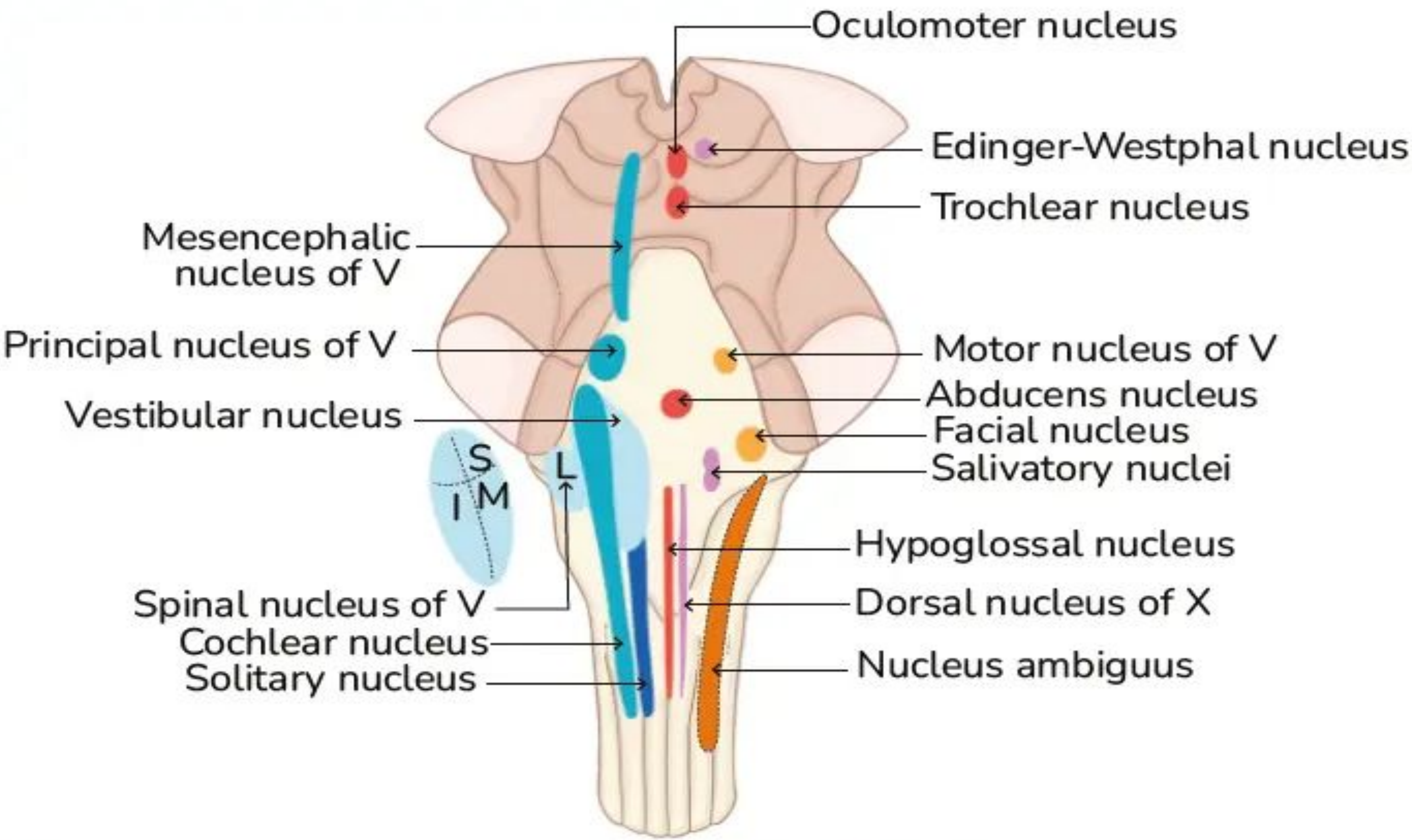
2. ***SPINAL NERVES***

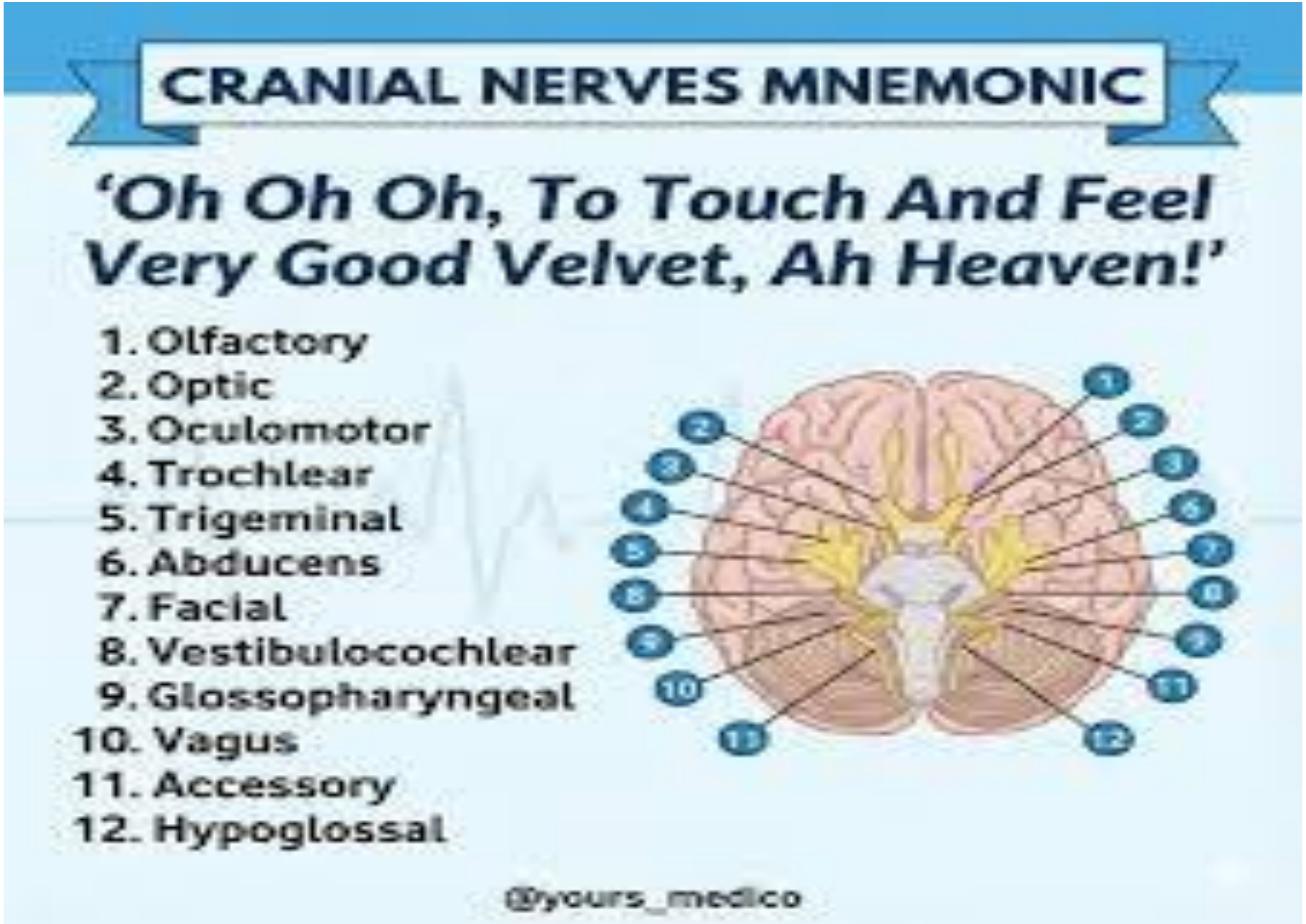
Spinal nerves are the 31 pairs of nerves branching from the spinal cord, responsible for sending motor commands and receiving sensory data from the body.

CRANIAL NERVES

- Cranial nerves are the 12 nerves of peripheral nervous system(PNS) that emerge from the foramina & fissures of the cranium.
- Their numerical order(1-12) is determined by their skull exit location(rostral to caudal).
- All cranial nerves originate from nuclei in the brain.
- Two originate from the forebrain(olfactory & optic), one has a nucleus in the spinal cord(accessory) while the remainder originate from the brain stem.


Olfactory I
Optic II
Oculomotor III
Trochlear IV
Trigeminal V
Abducens VI
Facial VII
Vestibulocochlear VIII
Glossopharyngeal IX
Hypoglossal XII
Vagus X
Accessory XI

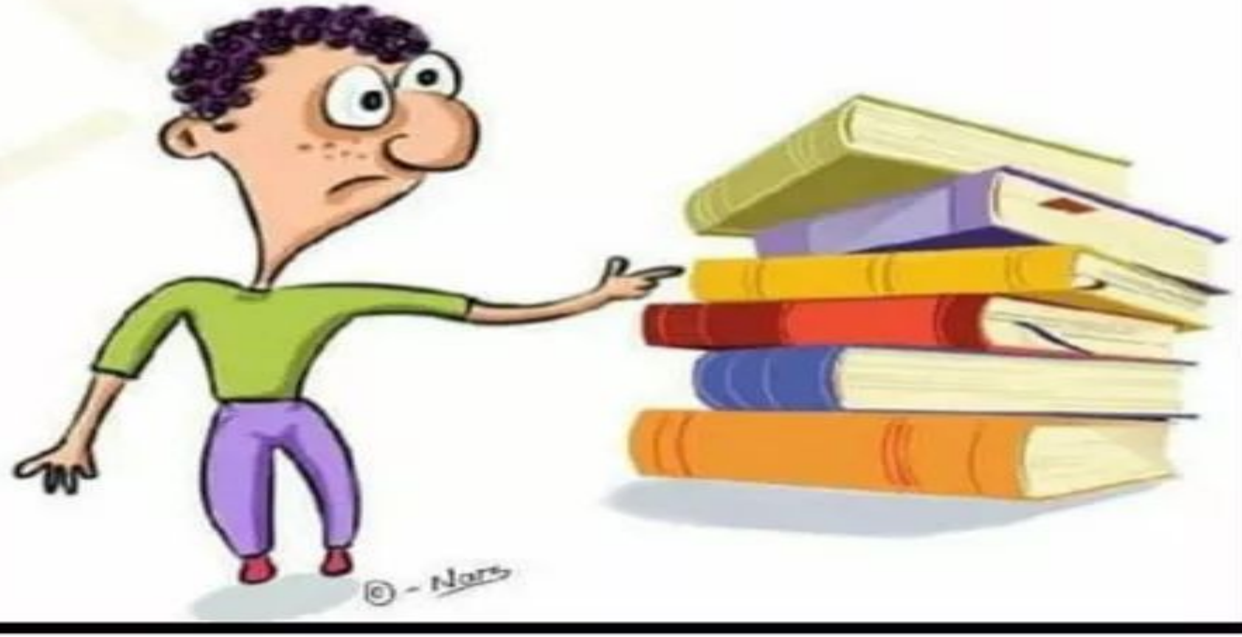

Oculomoter nucleus
Edinger-Westphal nucleus
Trochlear nucleus
Mesencephalic nucleus of V
Principal nucleus of V
Motor nucleus of V
Abducens nucleus
Vestibular nucleus
Facial nucleus
Salivatory nuclei
S
L
I
M
Hypoglossal nucleus
Spinal nucleus of V
Dorsal nucleus of X
Cochlear nucleus
Solitary nucleus
Nucleus ambiguus


CRANIAL NERVES MNEMONIC
'Oh Oh Oh, To Touch And Feel Very Good Velvet, Ah Heaven!'
1. Olfactory
2. Optic
3. Oculomotor
4. Trochlear
5. Trigeminal
6. Abducens
7. Facial
8. Vestibulocochlear
9. Glossopharyngeal
10. Vagus
11. Accessory
12. Hypoglossal
@yours_medico

Cranial nerve Types

	Cranial nerve
(S)ome	1
(S)ays	2
(M)oney	3
(M)atters	4
(B)ut	5
(M)y	6
(B)rother	7
(S)ays	8
(B)ig	9
(B)ooks	10
(M)atter	11
(M)ost	12

S = Sensory
M = Motor
B = Both



LESIONS OF CRANIAL NERVES

Cranial nerve lesion occur when any of the 12 pairs of nerves emerging from the brain are damaged by trauma, vascular issues, tumors or inflammation. As these nerves control specialized sensory and motor functions for the head & neck region, a deficit in any single nerve produces specific often ipsilateral(same side) effects.

COMMON CAUSES OF CRANIAL NERVE LESION

Cranial nerve lesions can be categorized by their etiology or location along the nerve pathway.

- ***VASCULAR ISSUES***

Ischemic strokes, aneurysms(e.g. PCOM aneurysm compressing CN-lll) or microvascular damage from diabetes.

CONT..

- ***TRAUMA***

Skull base fractures, whiplash or traumatic brain injury(TBI) can shear or compress nerve fibres.

- ***TUMORS***

Acoustic neuromas(affecting CN-Vlll) or meningiomas can exert pressure on nerves as they exit the skull.

- ***INFLAMMATION/INFECTION***

Cranial nerve lesion due to inflammation or cranial neuritis involve inflammatory damage, demyelination or ischemia causing dysfunction of one or more cranial nerves. Main causes are infections(VZV, TB, Lyme disease)& auto immune diseases(vasculitis, Tolosa Hunt syndrome).

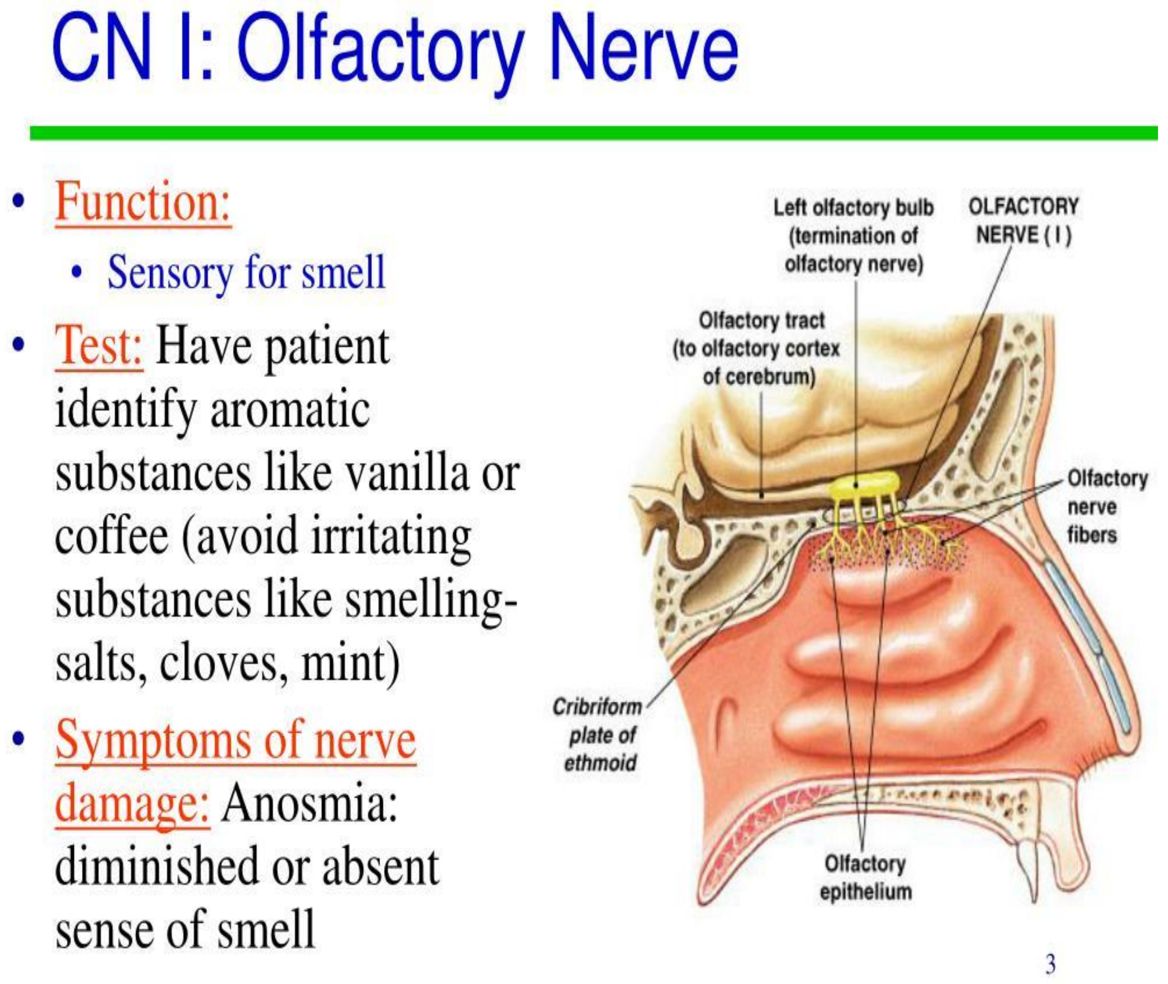
CRANIAL NERVES LESION & ASSOCIATED DEFICIT

- ***CRANIAL NERVE-I(OLFACTORY NERVE)***

Function: smell

Lesion deficit: **anosmia**(loss of smell)

Cause: head trauma(e.g cribriform plate damage), infection, tumor.



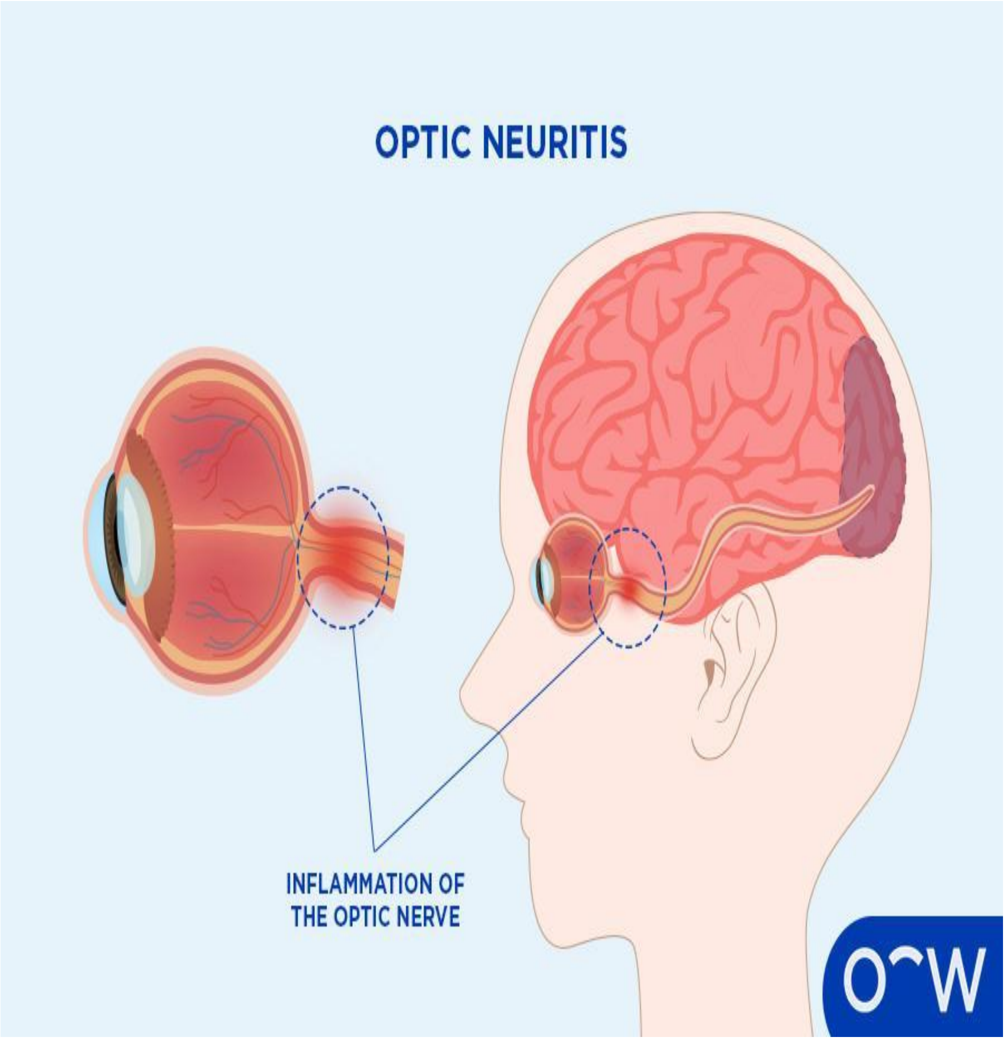
CONT..

- ***CRANIAL NERVE-II(OPTIC NERVE)***

Function: vision

Lesion deficit: blindness(complete lesion), visual field defects e.g. loss of peripheral vision(partial lesion)

Clinical example: optic neuritis.



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- ***CRANIAL NERVE-III(OCCULOMOTOR NERVE)***

Function: eye movement, pupil constriction

Lesion deficit: **ptosis**(drooping eyelid), dilated pupil, **eye deviation**(down & out) due to unopposed muscles

Cause: diabetes, aneurysm.

CONT..

- ***CRANIAL NERVE-IV(TROCHLEAR NERVE)***

 It is the thinnest nerve & can be damaged easily.

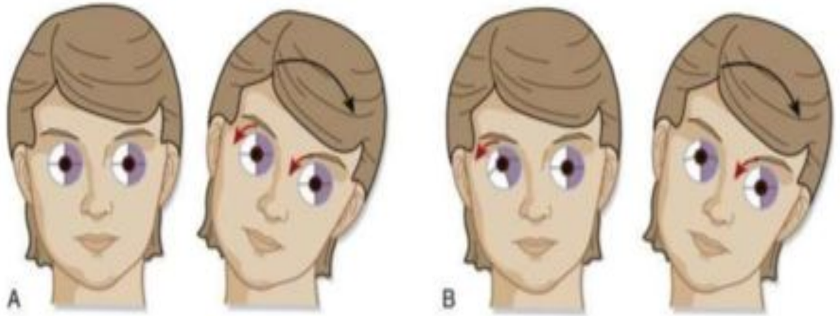
Function: downward eye movement(superior oblique muscle)

Lesion deficit: **vertical diplopia**(double vision when looking down e.g. walking downstairs)

Clinical presentation: patient tilts head away from the affected side to compensate.

Head tilt in 4th nerve palsy

- Superior oblique is an intorter
- When it's paralyzed, the eye becomes extorted by the unopposed action of the ipsilateral inferior oblique
- Patients will tilt their head to improve diplopia
- Example: Superior oblique palsy right eye, right eye i extorted, patient tilts left to fix the diplopia



Normal eye rotation
When the head tilts to the left, both eyes rotate in the opposite direction (right eye extorts, left eye intorts)

Cranial nerve IV palsy (right eye)
Right eye extorted and slightly elevated causing double vision. To compensate, the patient tilts her head to the left

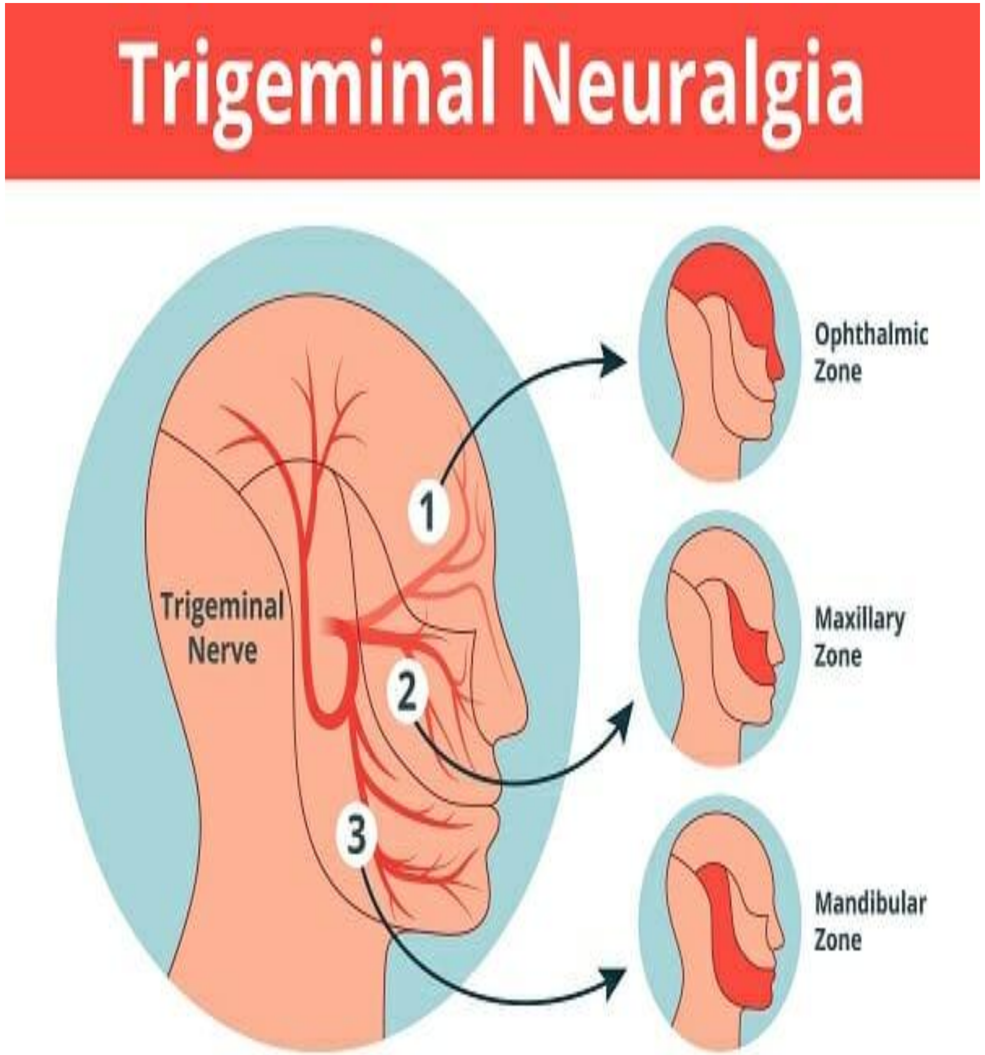
CONT..

- ***CRANIAL NERVE-V(TRIGEMINAL NERVE)***

 Trigeminal nerve is the largest cranial nerve & connects the brainstem to the face.

Function

it is major sensory nerve of face & also control motor functions like biting & chewing.



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Branches

Trigeminal nerve has three branches, the ***ophthalmic***(V_1), ***maxillary***(V_2) & ***mandibular***(V_3) that transmit sensation from the upper, middle & lower face to the brain.

TRIGEMINAL NEURALGIA

Symptoms

Intense, shooting, stabbing, electrical shock-like pain that lasts from seconds to minutes.

Location

Usually unilateral(one side of face).

Triggering factors

Light touch, shaving, brushing teeth, drinking, talking or a light breeze.

CONT..

Causes

When the nerve is injured or compressed typically at the base of brain e.g. neurovascular compression, multiple sclerosis(MS), tumors, stroke or facial trauma.

Treatment

- First line treatment usually includes anticonvulsants(carbamazepine or oxcarbazepine)to control nerve pain.
- If medication fails then surgical options like microvascular decompression, gamma-knife radiosurgery or nerve blocks.

CONT..

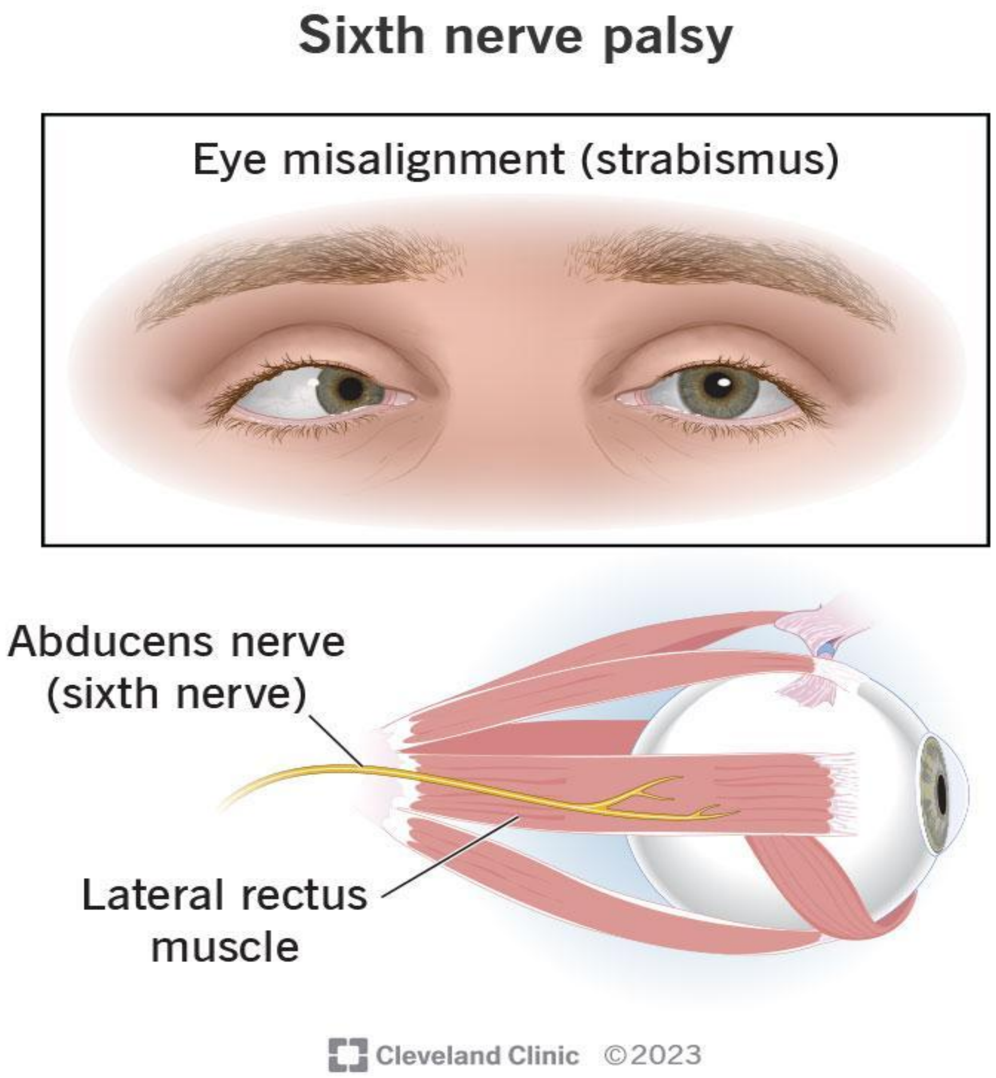
- ***CRANIAL NERVE-VI(ABDUCENS)***

Function: lateral eye movement i.e. moving eye outward(lateral rectus muscle)

Lesion deficit: inability to abduct(move outward) eye

Clinical presentation: **crossed eye**(medial deviation or eye pulled inward)

Cause: increased intracranial pressure.



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- ***CRANIAL NERVE-VII(FACIAL NERVE)***

Branches

Facial nerve is the seventh cranial nerve & travels a complex course through the temporal bone to reach its destination.Key branches of facial nerve include ***temporal***, ***zygomatic***, ***buccal***, ***mandibular*** & ***cervical***.

Function: Facial nerve is a ***mixed nerve*** controlling essential functions in the head & neck including ***facial expressions***(blinking, smiling, frowning or raising eye brows), ***taste*** from ant. 2/3 of tongue & ***parasympathetic control*** of lacrimal glands(tears), submandibular & sublingual glands(saliva) and mucous membranes in the nose/palate.

CONT..

It also provides sensation to a small area around the concha of the ear & supplies ***stapedius*** muscle in the middle ear protecting hearing by dampening loud noises.

BELL'S PALSY

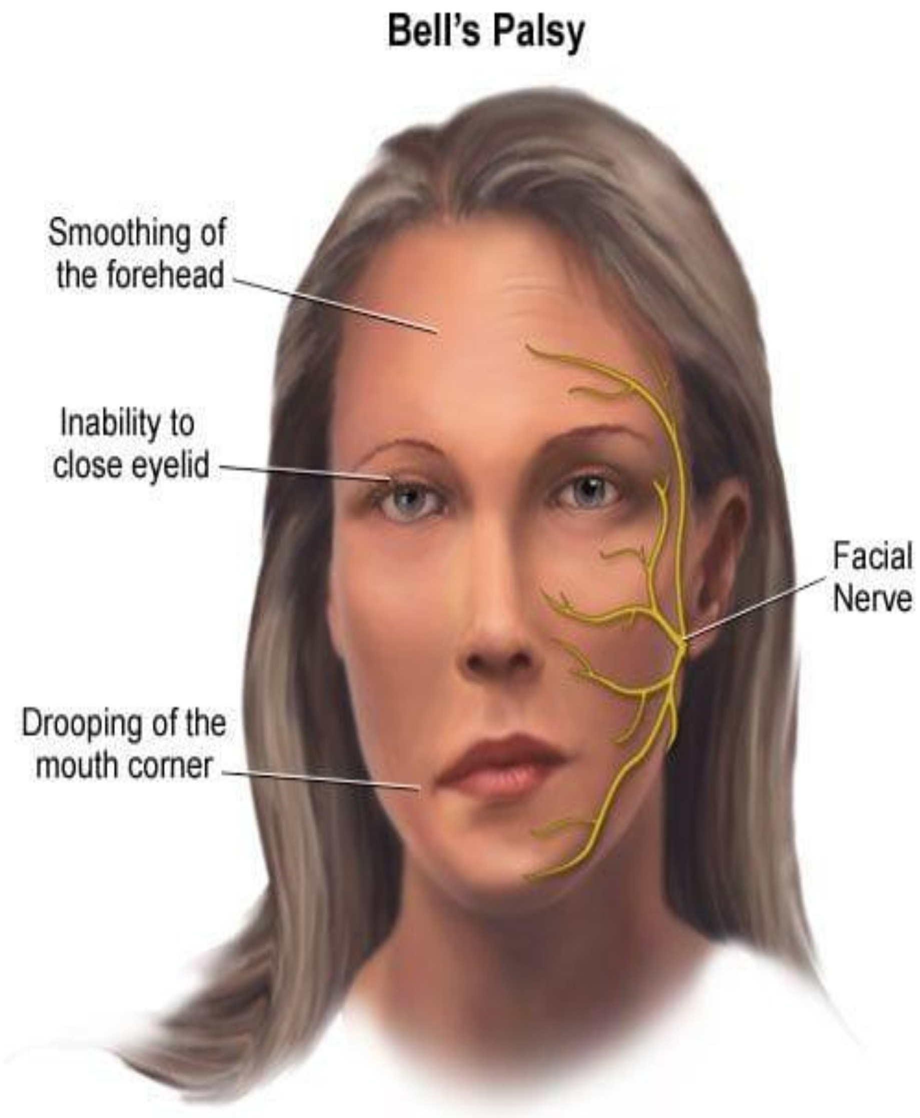
It is a sudden weakness or paralysis of facial muscles caused by the inflammation of seventh cranial nerve.

Symptoms

Typically affect one side of face causing drooping of mouth corners causing drooling and difficulty closing the eye lid leading to eye dryness.

Symptoms include pain behind the ear, headache & loss of taste on the front of the tongue.

It also causes increased sensitivity to sound on the affected side.



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Causes

The most common cause of bell’s palsy is the viral infection specially the ***herpes simplex virus***(causes cold sores), ***varicella zoster*** virus(chickenpox/shingles)or the ***Epstein barr*** virus.

The facial nerve becomes swollen & inflamed passing through a narrow bony passage which causes nerve compression.

CONT..

- ***CRANIAL NERVE-VIII(VESTIBULOCOCHLEAR NERVE)***

Function: hearing(cochlear nerve), balance(vestibular nerve)

Lesion deficit: loss of hearing, vertigo, dizziness(disequilibrium)

Cause: **acoustic neuroma** compressing the nerve.

CONT..

- ***CRANIAL NERVE-IX(GLOSSOPHARYNGEAL NERVE)***

It is the nineth cranial nerve originating from the post-olivary sulcus of the medulla oblongata in the brain stem & exits the cranium via jugular foramen.

Function:

- Taste, touch, pain and temperature sensation from posterior 1/3 of tongue, upper throat(pharynx), tonsills, middle ear.
- Motor functions include innervation of stylopharyngeus muscle to assist with swallowing & speech.

CONT..

Lesion deficit: loss of gag reflex, mild dysphagia(difficulty in swallowing), reduced taste on posterior 1/3rd of tongue, reduced saliva production(xerostomia) & glossopharyngeal neuralgia(a rare condition that cause severe pain in the throat, neck or tongue).

CONT..

- ***CRANIAL NERVE-X(VAGUS NERVE)***

Vagus nerve is the largest cranial nerve extending from the brainstem to the neck, chest & abdomen

Branches

Key branches include ***pharyngeal***(throat muscles), ***superior laryngeal***(larynx sensation), ***recurrent laryngeal***(voice box) & ***cardiac*** branches.

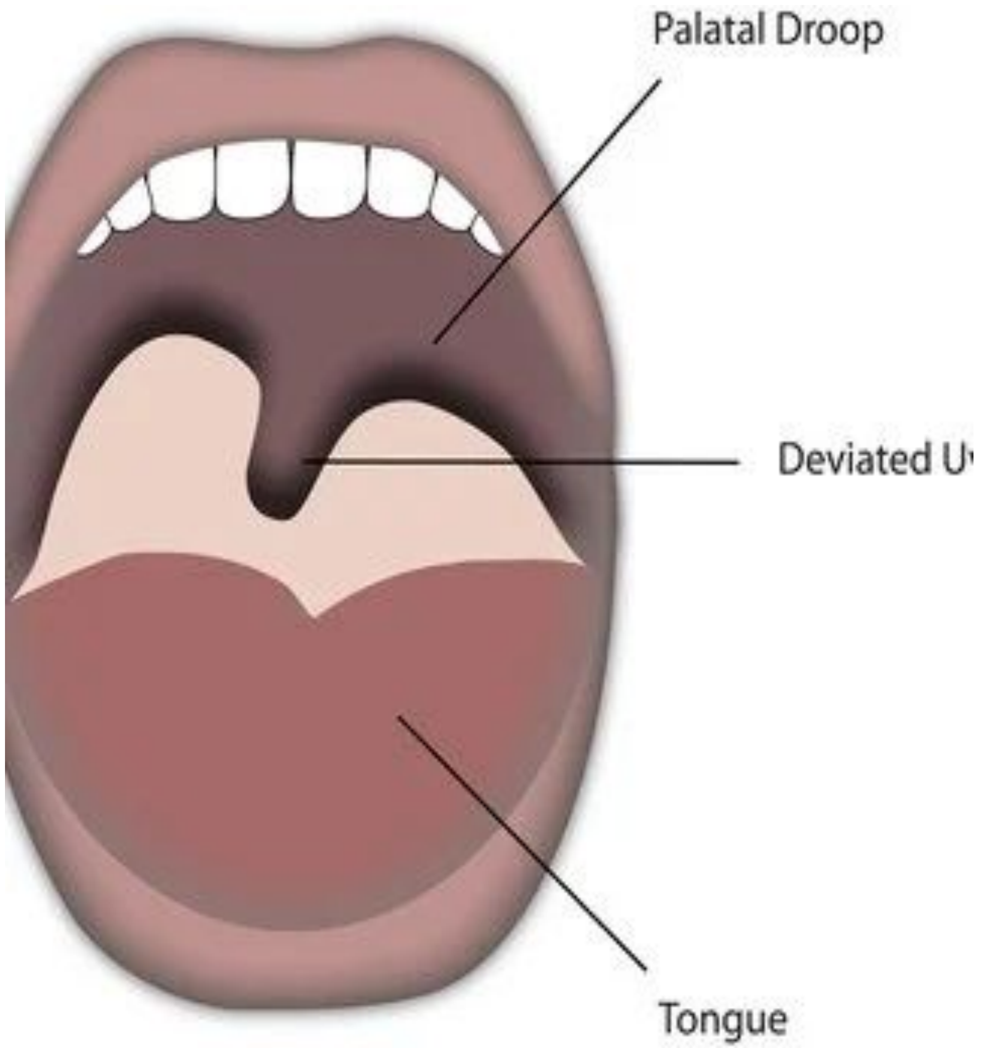
Function: parasympathetic control of heart & gut, digestion, voice, swallowing(larynx) & mood often acting as a brain-body communication highway.

Clinical importance: it has wide systemic importance(vital for autonomic control).

CONT..

Lesion deficit

Vagus nerve lesion cause significant dysfunction in parasympathetic regulation including dysphagia(difficulty in swallowing), hoarseness of voice, loss of gag reflex, uvula deviation towards the unaffected side, ipsilateral vocal cord paralysis & gastrointestinal issues(gastroparesis).



CONT..

Causes

- Surgical trauma from procedures like carotid endarterectomy, thyroid surgery or skull base surgery can injure the nerve.
- trauma from neck injuries or skull base fractures.
- Tumors and compression(structural issues in the cervical spine cause stretching).

Management

Treatment is largely symptomatic focusing on management of digestive issues, speech therapy & in some cases surgical voice rehabilitation.

CONT..

- ***CRANIAL NERVE-XI(ACCESSORY NERVE)***

Function: shoulder movement(trapezius), neck movement i.e. turning of head(sternocleidomastoid)

Lesion deficit: weak shoulder shrug, difficulty in turning head to opposite side.

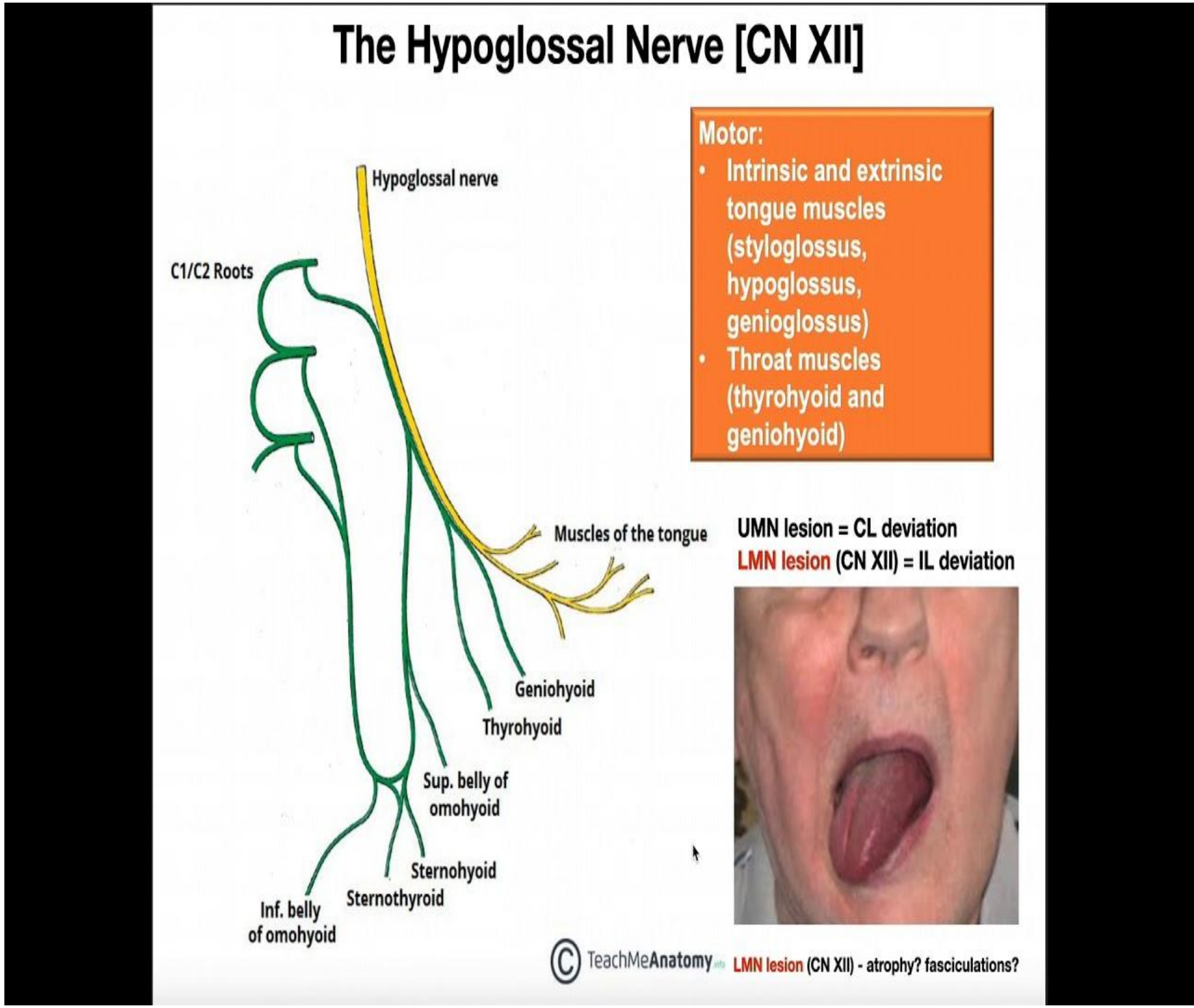
CONT..

- ***CRANIAL NERVE-XII(HYPOGLOSSAL NERVE)***

Function: movement of tongue

Lesion deficit: tongue deviates towards lesion, speech difficulty.

Clinical examination: ask patient to protrude the tongue.



EXAMINATION OF CRANIAL NERVES

- Smell test(CN-I)
- Visual acuity & fields test(CN-II)
- Eye movements(CN-III, IV, VI)
- Facial sensation & chewing(CN-V)
- Facial symmetry(CN-VII)
- Hearing test(CN-VIII)
- Gag reflex(CN-IX, X)
- Shoulder shrug(CN-XI)
- Tongue movement(CN-XII)

Table 1. **Functions and dysfunctions of the cranial nerves**

Cranial nerve name (number)	Type	Function	Associated dysfunction(s)
Olfactory (I)	Sensory	Sense of smell	• Unilateral or bilateral loss of sense of smell • Loss of taste
Optic (II)	Sensory	Vision	• Loss of vision
Oculomotor (III)	Motor	Movement of the eyeball and upper eyelid	• Eye-movement problems
	Parasympathetic	Pupil constriction	
Trochlear (IV)	Motor	Movement of the eyeball	• Eye-movement problems
Trigeminal (V)	Sensory	General sensation in face, scalp, corneas, and nasal and oral cavities	• Loss of facial sensation
	Motor	Chewing	
Abducens (VI)	Motor	Movement of the eyeball	• Eye-movement problems
Facial (VII)	Sensory	Taste	• Loss of taste • Inability to close eye
	Motor	Facial expression	
	Parasympathetic	Secretion of tears and saliva	
Vestibulocochlear (VIII)	Sensory	Hearing and balance	• Loss of hearing and balance
Glossopharyngeal (IX)	Sensory	Taste and sensation from back of tongue	• Inability to swallow • Hoarse voice
	Motor	Swallowing and speech	
	Parasympathetic	Secretion of saliva	
Vagus (X)	Sensory	Taste and sensation from epiglottis and pharynx	• Inability to swallow • Hoarse voice • Delayed gastric emptying
	Motor	Swallowing and speech	
	Parasympathetic	Muscle contraction of thoracic and abdominal organs and secretion of digestive fluids	
Accessory (XI)	Motor	Head and shoulder movement	• Inability to move head and raise shoulders
Hypoglossal (XII)	Motor	Movement of the tongue muscles	• Inability to move tongue

Source: Bayram-Weston (2020)

POST ASSESSMENT

Q1. A patient presents with complete loss of smell following a head injury. Which cranial nerve is most likely damaged:

a) Optic
b) Olfactory
c) Trigeminal
d) Facial

Q2. A patient has facial asymmetry with inability to close eye & drooping of mouth. This clinical picture suggests:

a) Trigeminal nerve lesion
b) Hypoglossal nerve lesion
c) Vagus nerve lesion
d) Facial nerve lesion

CONT..

Q3. A patient has difficulty shrugging shoulders & turning head to opposite side. Which nerve is affected:

a) Vestibulocochlear
b) Vagus
c) Accessory
d) Facial

Q4. Loss of taste from anterior two third of tongue is due to lesion of:

a) Trigeminal
b) Facial
c) Glossopharyngeal
d) Hypoglossal

Thank You!